

Sign and Conjugate Errors with Complex Numbers

Answer Key

Algebra 2

Quick Answers

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|---------------|---------------------|
| 1. $-2 - 6i$ | 5. $1 + 2i$ |
| 2. $23 - 14i$ | 6. $10, 6 - 2i$ |
| 3. 29 | 7. $3 + 2i, 3 - 2i$ |
| 4. $40 - 42i$ | 8. $2, -2, 2i, -2i$ |
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Curriculum

Level: Algebra 2

HSN-CN.A.1–A.3 – problems 1, 2, 3, 4, 5, 6

HSN-CN.C.7–C.9 – problems 7, 8

Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

3. *If they got 21: treated i squared as $+1$, so the product collapsed to 25 minus 4*
4. *If they got 58: used the conjugate product a squared plus b squared instead of squaring the same factor twice*
6. *If they got 8: subtracted the squares instead of adding them, as if i squared were $+1$*
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1. Subtract: $(3 - 8i) - (5 - 2i)$. The subtraction sign belongs to the whole second number, so distribute it to BOTH parts: $-(5 - 2i) = -5 + 2i$. Real parts: $3 - 5 = -2$. Imaginary parts: $-8i + 2i = -6i$. The result is $\boxed{-2 - 6i}$.

2. Multiply: $(4 + 3i)(2 - 5i)$. FOIL first: $8 - 20i + 6i - 15i^2$. Combine the middle terms: $8 - 14i - 15i^2$. Now use $i^2 = -1$, so $-15i^2 = +15$ — this is the sign students lose most often. $8 + 15 = 23$, and the answer is $\boxed{23 - 14i}$.

3. Conjugate pair: $(5 + 2i)(5 - 2i)$. The outer and inner products are $-10i$ and $+10i$, which cancel — that is why a conjugate pair has no i left. What remains is $25 - 4i^2$, and $-4i^2 = +4$ because $i^2 = -1$. $25 + 4 = 29$, so the product is $\boxed{29}$. Note the sign: the answer is $25 + 4$, not $25 - 4$.

4. Find the mistake (Nadia). (a) The conjugate of $7 - 3i$ is $7 + 3i$, not $7 - 3i$ again. Nadia multiplied the number by ITSELF and then reported the answer to a different product, $(7 - 3i)(7 + 3i) = 58$. To clear the i she must multiply by $7 + 3i$; only a conjugate PAIR makes the imaginary terms cancel. [Manual review: any explanation naming the conjugate $7 + 3i$ earns full credit.]

(b) Squaring the same factor: $(7 - 3i)(7 - 3i) = 49 - 21i - 21i + 9i^2$. Combine: $49 - 42i + 9i^2$, and $9i^2 = -9$. So $49 - 9 = 40$ and the correct value is $\boxed{40 - 42i}$ — still imaginary, which is exactly the evidence that the wrong factor was used.

5. Divide: $\frac{8+i}{2-3i}$. Multiply numerator and denominator by the conjugate of the denominator, $2 + 3i$. Numerator: $(8+i)(2+3i) = 16 + 24i + 2i + 3i^2 = 16 + 26i - 3 = 13 + 26i$. Denominator: $(2-3i)(2+3i) = 4 + 9 = 13$ (a sum, because $-9i^2 = +9$). Divide each part by 13: $\frac{13+26i}{13} = \boxed{1+2i}$.

6. Find the mistake (Marcus). (a) Marcus computed $(3+i)(3-i)$ as $9 - 1 = 8$, which treats i^2 as $+1$. The middle terms cancel and the last term is $-i^2 = +1$, so the product is $9 + 1$, and the correct denominator is $\boxed{10}$. [Manual review: the explanation must say the last term flips sign because $i^2 = -1$.]

(b) Numerator: $20(3-i) = 60 - 20i$. Divide by the correct denominator 10: $\frac{60-20i}{10} = \boxed{6-2i}$. Check by multiplying back: $(6-2i)(3+i) = 18 + 6i - 6i - 2i^2 = 18 + 2 = 20$.

7. Solve: $x^2 - 6x + 13 = 0$. Discriminant: $(-6)^2 - 4(1)(13) = 36 - 52 = -16$, negative, so the two solutions are non-real. $\sqrt{-16} = 4i$. Quadratic formula: $x = \frac{6 \pm 4i}{2} = 3 \pm 2i$. The two solutions are $\boxed{x = 3 + 2i}$ and $\boxed{x = 3 - 2i}$. They are conjugates of each other: same real part, opposite signs on the imaginary part. That always happens when a quadratic with real coefficients has a negative discriminant.

8. All zeros of $P(x) = x^4 - 16$. Factor as a difference of squares twice: $x^4 - 16 = (x^2 - 4)(x^2 + 4)$, and $x^2 - 4 = (x - 2)(x + 2)$. From $x^2 - 4 = 0$: $\boxed{x = 2}$ and $\boxed{x = -2}$. From $x^2 + 4 = 0$: $x^2 = -4$, so $x = \pm\sqrt{-4}$, giving $\boxed{x = 2i}$ and $\boxed{x = -2i}$. The sign trap is $x^2 + 4$: it does NOT factor over the reals, and its roots are $\pm 2i$, a conjugate pair — writing only $2i$ leaves a fourth-degree polynomial with three zeros.